Rebuttal: Anti-correlated Noise Provably Ensures Stable Convergence to Flat Minima

	CIFAR-10N			CIFAR-100N
	Random1 (Nr ~ 0.09)	Aggregate (Nr ~ 0.17)	Worst(Nr ~ 0.4)	Noisy(Nr ~ 0.4)
Anti-SGD	81.83 ± 0.21	88.30 ± 0.22	62.79 ± 0.40	54.29 ± 0.27
cANI	$\mathbf{88.67 \pm 0.21}$	$\mathbf{90.34 \pm 0.16}$	$\mathbf{81.50 \pm 0.32}$	$\mathbf{58.38 \pm 0.32}$
PSGD	40.74 ± 1.35	43.15 ± 0.54	33.98 ± 1.06	31.66 ± 0.23
SGD	84.51 ± 0.36	89.83 ± 0.33	66.38 ± 0.52	54.28 ± 0.40

TABLE R.1. Test accuracy (%) on test datasets with real-world datasets CIFAR-10N/100N. Results are presented as the average and the standard deviation. The anti and pgd methods use the same perturbation scale σ, while cANI uses ($\sigma = 0.1,\ \tau = 0.05$) for CIFAR-10N and ($\sigma = 0.03,\ \tau = 0.5$) for CIFAR-100N.

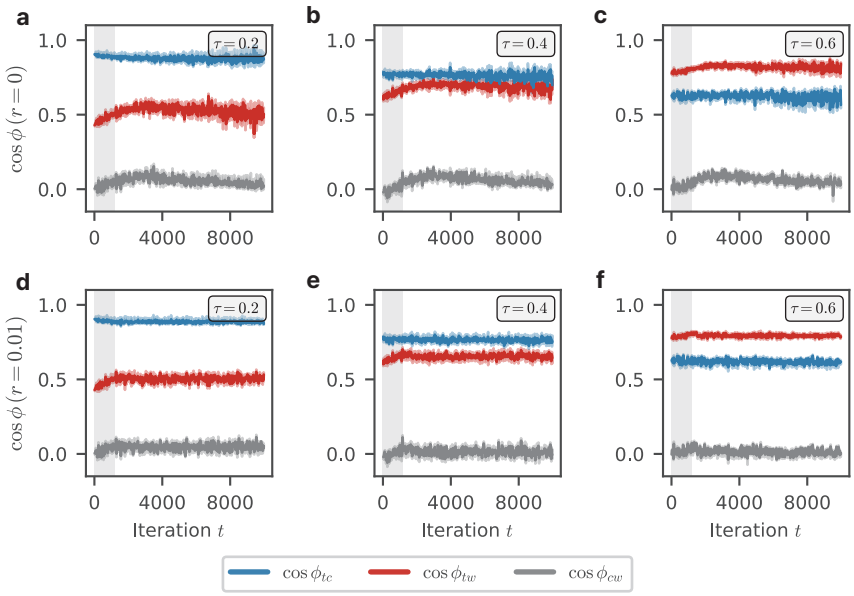

Figure R.1. Cosine similarities between $-\boldsymbol{\nabla}_{\boldsymbol{\theta}}\mathcal{R}_{\tilde{\mathcal{D}}_{\text{tr}}}(\boldsymbol{\theta})$ (total drift), $-\boldsymbol{\nabla}_{\boldsymbol{\theta}}\hat{\mathcal{R}}_{\tilde{\mathcal{D}}^c_{\text{tr}}}(\boldsymbol{\theta})$ (drift from the correct label, i.e., correct drift), and $-\boldsymbol{\nabla}_{\boldsymbol{\theta}}\hat{\mathcal{R}}_{\tilde{\mathcal{D}}^w_{\text{tr}}}(\boldsymbol{\theta})$ (drift from the wrong label, i.e., wrong drift). The (a, b, c) first row shows the cosine similarities when there is no restart $r = 0$, and the (d, e, f) second row shows the cosine similarities when the restarting is applied $r = 0.01$, with (a, d) the noise rate $\tau = 0.2$, (b, e) $\tau = 0.4$, and (c, f) $\tau = 0.6$. Here, $\cos\phi_{tc}$, $\cos\phi_{tw}$, and $\cos\phi_{cw}$ denote the cosine similarity between total and correct drifts, total and wrong drifts, and correct and wrong drifts, respectively. The shaded gray area represents the iterations before the restarting commences, which portrays the region where the dynamics should remain the same regardless of the restart probability r. We set $B = 8$ in Setting 1. We performed 5 independent runs, where the shaded areas denote the standard error.

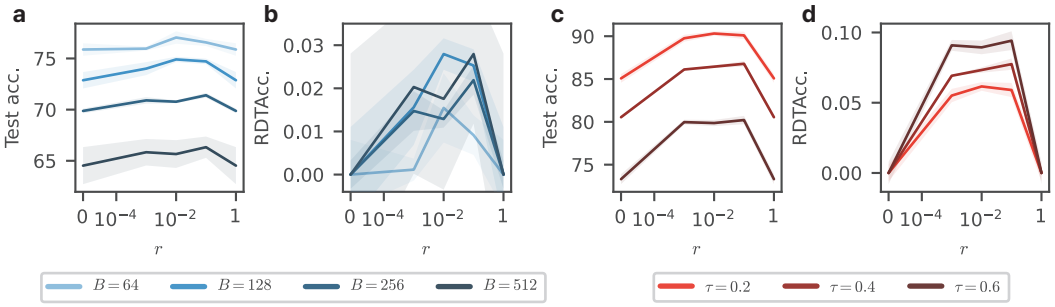

Figure R.2. Test accuracy (Test acc., left) and relative difference of test accuracy (RDTAcc., right) results with (a, b) varying the batch size B, and (c, d) varying the noise rate τ with respect to the restart probability r. We use CIFAR-10 dataset and CE loss following the settings in Table 1. For (a, b), we used noise rate $\tau = 0.6$, fixed learning rate 0.02. We make $t_f - t_m$ relatively equal across batch size where t_f is the iteration where the model is chosen to evaluate the metric, and t_m is the overfitting iteration. For (c, d), we used the batch size $B = 256$ as following the settings in Table 1. We performed 3 independent runs, where the shaded areas denote the standard error.